KARNAVATI UNIVERSITY | UIT

NAAC A+ ACCREDITED

AI-Based Network Intrusion Detection System

Course: Computer Networks | Course Code: _____

FACULTY

Ms. Pragya Vats

GROUP MEMBERS

Darshan Gujjar (KU2407U039) | Panchal Shiv (KU2407U201) | Aryan Panchal (KU2407U025)

CLIENT — ROUTER — AI / IDS — FIREWALL / SERVER

INTERACTIVE Q&A

Discussion & Revision

1. What is an Intrusion Detection System (IDS)?
2. Why do we divide data into training and testing sets?
3. What is the difference between precision and recall?
4. What is a false positive in network intrusion detection?
5. Which Computer Networks syllabus unit is most directly related to this project?

PROJECT FOUNDATION

What is this project about?

PURPOSE

Automated Detection

Use machine learning algorithms to identify network traffic that may be suspicious or malicious.

NETWORKING LINK

Protocol Metrics

Leverages IP/flow information, protocols, packet counts, and communication traffic patterns.

EXPECTED OUTPUT

Real-Time Classification

Predicts Normal vs. Suspicious traffic and displays classified results via UI dashboard.

BACKGROUND CONTEXT

Introduction to Network Security

Computer Network

Devices exchange data using IP, TCP, UDP, DNS, and other fundamental transport protocols.

Security Challenge

Suspicious traffic can be difficult to manually identify across large enterprise network topologies.

AI Opportunity

AI models learn hidden patterns from historical examples to assist analysts reviewing traffic.

PROBLEM DEFINITION

Problem Statement

PROBLEM

Manual Bottleneck

Large volumes of live network traffic make manual packet inspection slow and prone to errors.

NEED

Automated Classifier

A dedicated system that examines flow features and flags malicious traffic flows automatically.

PROJECT SCOPE

Prototype Boundaries

A learning prototype only; generated alerts require human review and continuous validation.

PROJECT GOALS

Aim and Objectives

PRIMARY AIM

Build an AI/ML prototype for network intrusion detection.

OBJECTIVE 1

Data Preparation

Prepare network-flow data and analyze relevant domain features.

OBJECTIVE 2

Model Training

Train a classifier to distinguish normal from suspicious network traffic.

OBJECTIVE 3

Evaluation

Evaluate the classifier model and visualize overall system performance.

COMPARATIVE ANALYSIS

Existing System and Limitations

TRADITIONAL APPROACH

Static Signature Matching

Relies on known attack signatures, fixed rule sets, and manual packet inspection routines.

LIMITATIONS

Rule Brittleness

Rules require constant manual updates, and unfamiliar or zero-day attack patterns are missed.

BALANCED VIEW

Human-in-the-Loop AI

AI models can also make false errors and must support — rather than replace — security experts.

MACHINE-LEARNING WORKFLOW

Proposed AI-Based System

1

Input

A labeled dataset containing diverse network-flow features.

→

2

Preparation

Clean raw data, encode categorical labels, and form train/test splits.

→

3

Prediction

Train a model such as Random Forest to classify traffic.

→

4

Output

Show predictions, evaluation scores, and flagged packet logs.

ACADEMIC ALIGNMENT

Computer Networks Concepts Used

Unit I

Application protocols and network services monitoring.

Unit II

TCP/UDP, port analysis, flow duration, and packet counts.

Unit III

IP addressing, forwarding rules, and source/destination metadata.

Unit IV

Link-layer communication and packet analysis.

Unit V (Direct Project Relation)

Intrusion detection, operational security, and network monitoring tools.

HIGH-LEVEL DESIGN

System Architecture

MODULE 1

1. Data Selection

Use a benchmark public dataset or authorized lab traffic captures.

MODULE 2

2. Feature Processing

Select relevant flow fields and clean raw tabular data.

MODULE 3

3. AI Model

Train and test the classifier on separate isolated data splits.

MODULE 4

4. Dashboard UI

Display predictions, confusion matrix, and performance scores.

DATA SPECIFICATION

Dataset & Network Traffic Features

DATASET OPTION

CIC-IDS2017

A recognized public dataset; verify official documentation before deployment.

EXAMPLE FEATURES

Flow Metrics

Protocol, flow duration, packet totals, byte totals, lengths, and flow rates.

LABELS

Target Categories

Normal benign traffic alongside specific attack categories within dataset.

DATA QUALITY

Leakage Prevention

Ensure complete separation between training and test sets to prevent leakage.

METHODOLOGY

AI/ML Algorithm and Methodology

ALGORITHM

Random Forest

Combines multiple decision tree classifiers for robust network feature classification.

WORKFLOW

4-Step Sequence

Load & clean → train/test split → train model → predict and evaluate metrics.

BASELINE COMPARISON

Model Benchmarking

Compare ensemble performance against a single baseline Decision Tree model.

REPRODUCIBILITY

Experiment Logging

Record exact dataset version, feature selections, split seeds, and hyperparameters.

TECHNICAL STACK

Tools and Technologies

LANGUAGE

Python

Primary programming language for network data processing and model building.

INTERFACE

Streamlit / Jupyter

Interactive web UI dashboard or Jupyter Notebook report for visualization.

LIBRARIES

Core ML Libraries

Pandas, NumPy, Scikit-learn, and Matplotlib/Seaborn visual tools.

NETWORK ANALYSIS

Wireshark

Inspect authorized sample traffic flows while protecting private user payload data.

EXECUTION PIPELINE

Implementation and Working

Step 1 Load network flow dataset and check for missing or invalid values.

Step 2 Prepare feature matrices and target labels; encode categorical data if required.

Step 3 Split prepared dataset into dedicated training and testing subsets.

Step 4 Train the classifier model and predict traffic labels for unseen test data.

Step 5 Display sample traffic predictions and visual evaluation metrics on dashboard.

EVALUATION FRAMEWORK

Results and Performance Metrics

METRIC

Accuracy

Overall proportion of correct predictions across all evaluation traffic flows.

METRIC

Precision

Of flows flagged as attacks by the model, how many were true attacks?

METRIC

Recall

Of all actual attack flows, how many did the trained model detect?

METRIC

F1-Score

A combined harmonic metric measuring precision and recall balance.

Results Visual: Insert actual confusion matrix plot and recorded metrics table upon prototype completion.

PROJECT APPRAISAL

Advantages, Limitations & Applications

ADVANTAGES

Automated First-Pass

Automates first-pass log review and enables repeatable empirical testing.

LIMITATIONS

Dataset Dependency

Quality relies on data source; false positives and missed attacks can occur.

APPLICATIONS

Use Cases

Security learning labs, educational tools, and admin monitoring support.

ETHICS

Data Responsibility

Use authorized traffic datasets only and keep captured network data private.

PROJECT CLOSURE

Conclusion and Future Scope

Conclusion

The prototype successfully demonstrates how machine learning can support modern network traffic classification.

FUTURE SCOPE 1

Model Benchmarking

Compare multiple classifier algorithms and broader attack category detection.

FUTURE SCOPE 2

Explainability

Integrate model explainability (XAI) to highlight features driving alert predictions.

FUTURE SCOPE 3

Cross-Validation

Test on independent external datasets to evaluate false-positive performance.

ACADEMIC CITATIONS

References

DATASET CITATION

CIC-IDS2017

Canadian Institute for Cybersecurity (CIC), CIC-IDS2017 Dataset Documentation.

MACHINE LEARNING

Scikit-learn Guide

Scikit-learn User Guide: Random Forest Classifiers and Model Evaluation Metrics.

PACKET ANALYSIS

Wireshark Docs

Wireshark User's Guide and Protocol Dissection Standards Documentation.

FINAL REPORT NOTE

Citation Requirement

Add exact URLs, access dates, library version numbers, and consulted research papers.